STRATEGIC MARKET SELECTION & RISK-AWARE UNIT MIX OPTIMIZATION

SKYLINE CAPITAL PARTNERS – CLASS B DEVELOPMENT ANALYSIS

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INVESTMENT OBJECTIVE & ANALYTICAL FRAMEWORK

Identify	Identify the most attractive Class B development market.
Evaluate	Evaluate opportunities using risk-adjusted return analysis.
Optimize	Optimize unit mix under structural, cost, and demand constraints.
Control	Explicitly control downside risk exposure.
Market	Framework: Market Evaluation → Risk-Adjusted Return Analysis → Monte Carlo Simulation → Constrained Optimization → Investment Recommendation

MARKET COMPARISON: EXPECTED RETURNS

- New Hope delivers materially higher expected returns with superior downside performance.
- New Hope maintains the strongest downside IRR floor (P05 = 4.37%), reinforcing resilience in adverse scenarios.

City	Mean IRR	Std Dev IRR	P05 IRR
New Hope	**8.8%**	3.1%	**4.37%**
Fairview	5.9%	1.93%	3.29%
Springfield	5.08%	1.74%	2.96%

New Hope offers the highest expected IRR and strongest fifth-percentile return, despite greater return volatility.

RISK DRIVERS ACROSS MARKETS

Operating Risk Drivers:

Vacancy volatility broadly comparable across markets

Lowest operating expense ratio (New Hope)

Strongest average NOI performance (New Hope)

Highest downside IRR (P05)

Conclusion: New Hope's higher expected return is supported by stronger operating performance, although its IRR volatility is also higher.

MARKET SELECTION: NEW HOPE

New Hope is the preferred development market, with the highest estimated mean IRR (8.8%) and strongest fifth-percentile IRR (4.37%). Although return volatility is higher, its favorable operating performance and downside-return profile support the investment recommendation.

DEVELOPMENT CONSTRAINTS (MEMO-BASED)

These constraints define the feasible solution space for optimization.

Physical Constraints

- Fixed development capacity of 85 units
- Maximum development area of 80,000 sq ft

Cost Structure

- Fixed annual operating costs
- Unit-specific variable operating costs

Market Constraints

- Rent-dependent demand caps
- Vacancy uncertainty modeled using historical observations

Risk Constraints

- Maximum modeled loss frequency of 10%
- All constraints are embedded directly into the Monte Carlo optimization model.

MONTE CARLO RISK SIMULATION

Simulated rent distributions from historical Class B data.

Simulated vacancy using empirical distribution.

Applied demand caps by rent tier.

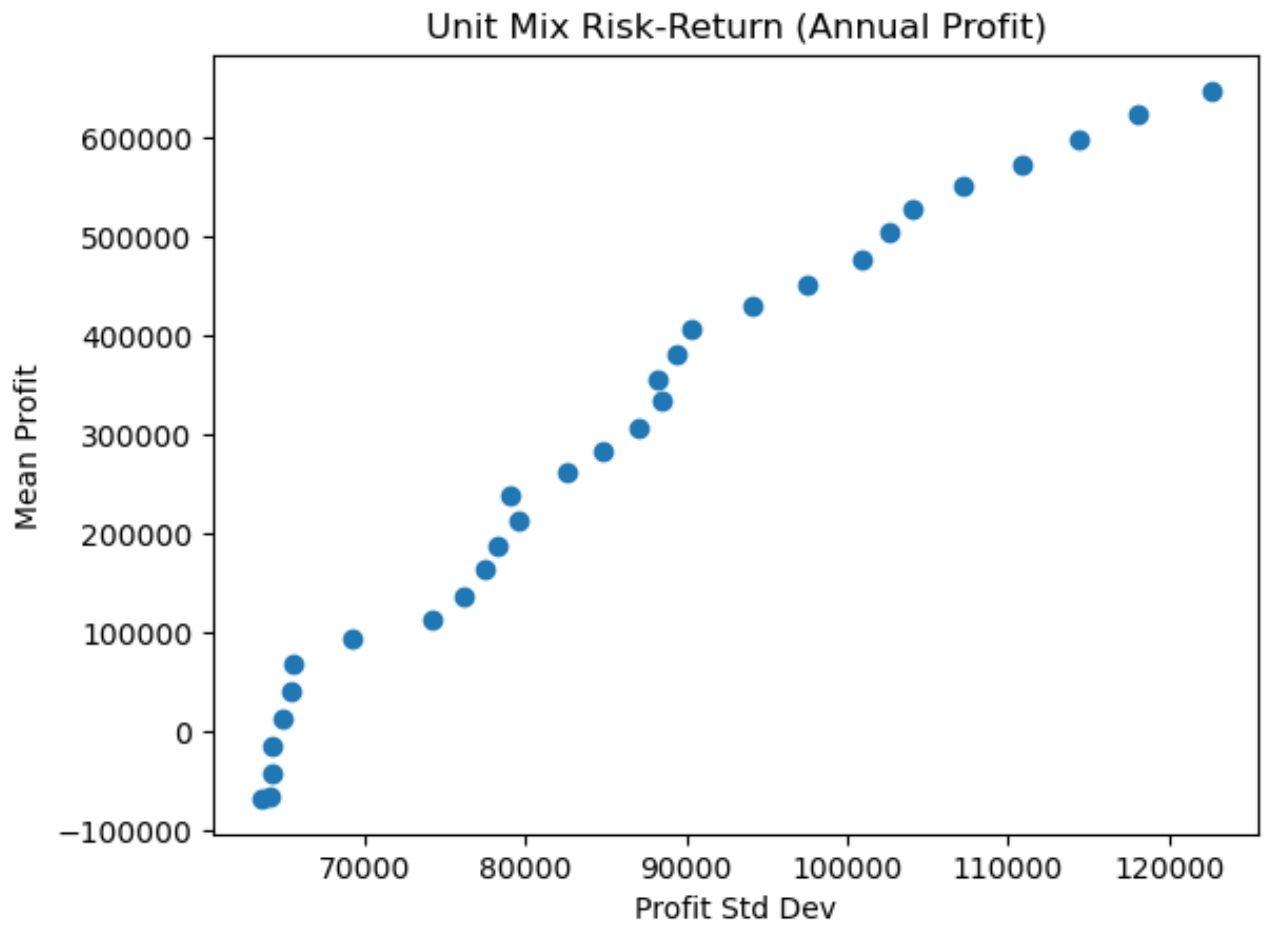
Calculated:

- Expected annual profit
- Profit volatility
- Probability of loss

20,000 Monte Carlo trials using empirical historical distributions

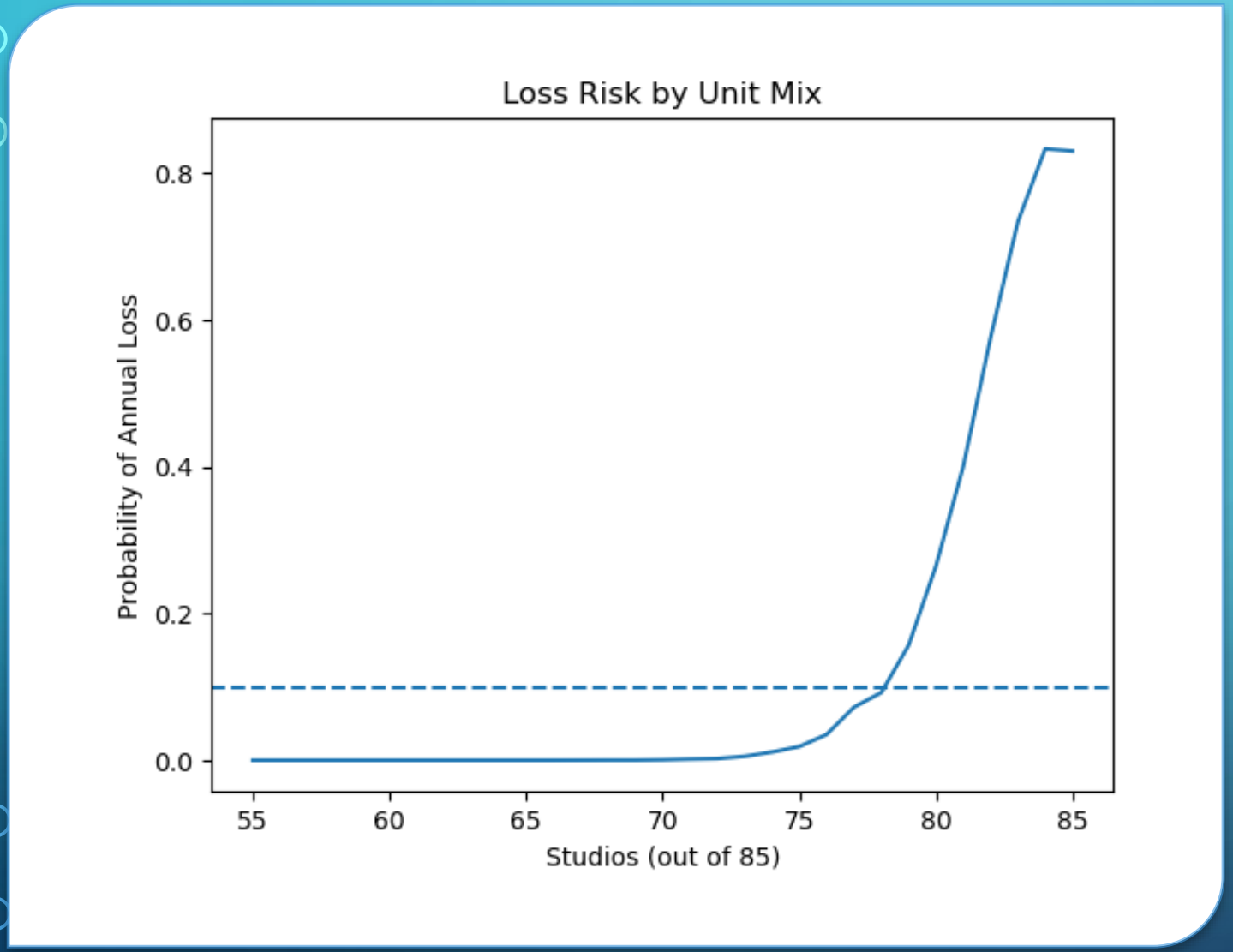
No parametric distributional assumptions imposed

Sample Rent and Vacancy → Apply Demand Caps → Calculate Occupancy and Revenue → Deduct Costs → Estimate Profit Distribution



RISK-RETURN TRADEOFF

- Feasible unit mixes exhibit a tradeoff between expected annual profit and profit volatility. Higher expected profit is generally associated with higher variability, making risk constraints important when selecting the development configuration.



DOWNSIDE RISK ANALYSIS

- Simulated loss frequency rises sharply at high studio concentrations. The 10% loss-frequency threshold identifies configurations that exceed the modeled risk tolerance.

OPTIMAL RISK-CONSTRAINED MIX

Recommended Mix: 55 Studios / 30 One-Bedrooms

Metrics:

Mean Profit: ~$645K

Std Dev: ~$122K

P05 Profit: ~$436K

Observed Simulated Loss Frequency: 0.00%

Full space utilization (80,000 sq ft)

Selected as the highest expected-profit configuration satisfying the modeled ≤10% loss-frequency constraint.

Note: No losses were observed across 20,000 simulated scenarios; actual losses remain possible.

RISK COMPARISON: TOP FEASIBLE UNIT MIXES (≤10% LOSS CONSTRAINT)

Studios	1BR	Mean Profit	Std Dev	P05 Profit	Loss Freq.
55	30	**$644,529**	$122,340	**$435,570**	0%
56	29	$621,327	$119,090	$419,426	0%
57	28	$597,278	$115,018	$401,431	0%

The recommended 55-studio / 30-one-bedroom configuration delivers the highest expected annual profit among the evaluated risk-compliant unit mixes.

Alternative configurations may offer lower profit volatility, but at the expense of expected returns and downside profitability.

The recommended mix provides a favorable balance between profitability and modeled risk exposure.

The preferred configuration maximizes expected profit within the modeled loss-frequency constraint.

STRESS TEST: +2 PERCENTAGE-POINT VACANCY

Metric	Baseline	+2% Vacancy
Mean Profit	$644,529	$603,347
Profit Std Dev	$122,340	$117,927
P05 Profit	$435,570	$399,554
Observed Loss Frequency	0%	0%

- A two-percentage-point increase in vacancy reduces expected annual profit by approximately $41,182 (6.4%).
- The fifth-percentile profit declines by approximately 8.3%, indicating weaker downside performance under the stress scenario.
- No losses were observed across 20,000 simulated stress scenarios; actual investment losses remain possible.

FINAL INVESTMENT RECOMMENDATION

Based on comprehensive return, volatility, and downside analysis:

Recommended Market:

- New Hope
- Estimated mean IRR: 8.8%

Recommended Unit Mix:

- 55 Studios / 30 One-Bedrooms
- Expected annual profit: ~$645K

- Highest expected IRR among evaluated markets
- Strongest fifth-percentile IRR (4.37%)
- Favorable modeled downside protection
- Meets the ≤10% simulated loss-frequency constraint
- Resilient under the +2 percentage-point vacancy stress scenario
- Full utilization of the available development space

The recommended strategy combines attractive expected returns with disciplined risk management and resilience under modeled stress conditions.

USE OF GENERATIVE AI

Generative AI tools, including ChatGPT, were used to support financial-model logic review, Monte Carlo simulation development, Python code refinement, and presentation clarity.

Analytical assumptions, calculations, simulation outputs, and investment recommendations were reviewed as part of the project workflow. AI-generated suggestions were evaluated before incorporation into the final analysis.

Portfolio adaptation of an academic analytics project. The original course-provided dataset is not publicly distributed. Results reflect modeled assumptions and simulated outcomes.